Tracing multi-religiosity and interethnic networks among Iranian communities in the Red Sea (19th-20th centuries)

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Abstract

This article examines the potential of Digital Humanities to trace the life trajectories of members of the Iranian communities in the Red Sea region between the 19th and 20th centuries, with a specific focus on multi-religiousness and the interethnic networks they established. Drawing upon official documents, personal accounts, and the Iranian press in Egypt, this research analyses the dynamics of encounter and mixing between Iranian subjects and local populations on both shores of the Red Sea. In doing so, it specifically focuses on the multiple religious belonging of these Iranian subjects: Shi'a and Sunni Muslims, Baha'is, Jews, and Christians, who moved from Iran to the Red Sea region during the 19th and 20th centuries (Yadegari, 1980; Field, 1985; Luesink, 1993), when diasporic and religious networks blurred and intersected (Fig. 1).

This article focuses on the methodological and epistemological challenges associated with the study of these fluid belongings and their representation through digital humanities tools. In this regard, the potential of two open-source tools is explored: Gephi for social network analysis (Bastian et al., 2009) and Palladio for the spatio-temporal visualisation of life trajectories (Conroy, 2021). Addressing the multilingual and multiscriptural nature of Arabic and Persian sources, the article explicitly links this challenge to current developments in Handwritten Text Recognition (HTR) (Broadwell et al., 2025) and Optical Character Recognition (OCR) (Buoy et al., 2023), evaluating how emerging models (e.g., Kraken, Transkribus) can partially overcome the limitations of tools designed primarily for the Latin alphabet (Buoy et al., 2023). The paper addresses two main critical issues. First, the semantics of identities (ethnic, religious, mixed), combined with the "silence" of sources, complicates the definition of categories and the assignment of metadata. Regarding network analysis, the methodological choices for determining node and edge weights are clarified and justified: node size reflects the documented population of Iranian origin in a given city, based on cross-referenced prosopographic data; edge thickness represents the strength of each religion's presence in a specific location, calculated as a composite index that combines frequency of attestation in primary sources, longevity of documented settlement, and intensity of interreligious interactions recorded in personal accounts and community press. These choices are deliberately grounded in source criticism rather than assuming unproblematic quantitative equivalence. Second, the multilingual and multiscriptural nature of Arabic and Persian sources poses significant technical challenges, highlighting the need for innovative solutions for tools designed

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primarily for the Latin alphabet, a need to which HTR/OCR technologies offer partial responses.

Despite these challenges, the article demonstrates how critical and informed use of these tools can contribute to writing more inclusive histories of Iranian communities in the Red Sea, reflecting the complexity of their inter-ethnic networks and multi-religious experience. The analysis reveals that primary sources systematically overrepresent religious minorities (Baha'is, Jews, Christians, and Sunni Muslims) while underrepresenting the majority Shia population, as also suggested by the large number of individuals whose religious belonging is not specified.

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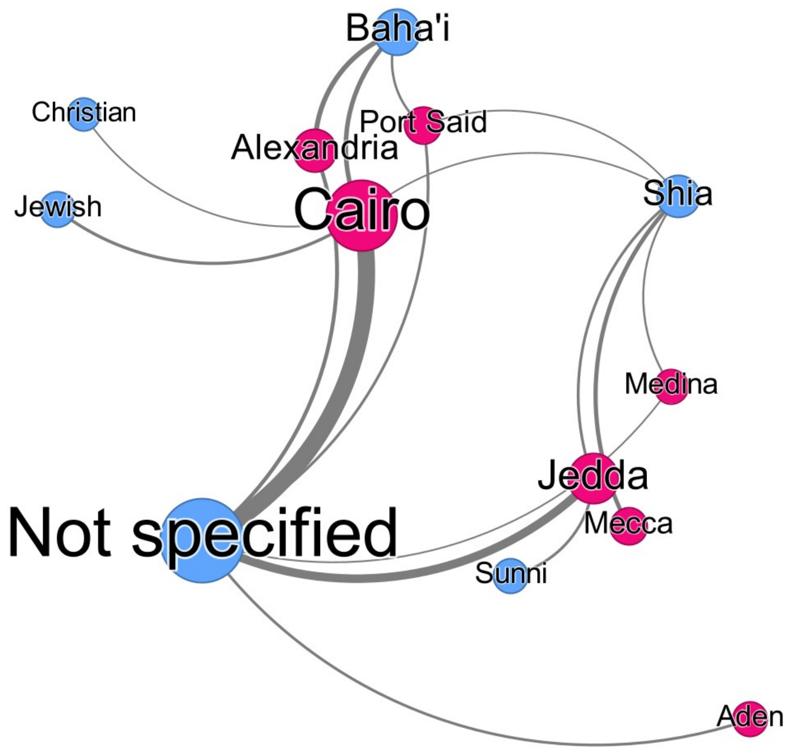

Figure 1. Religious groups among Iranians in the Red Sea, 19th and early 20th centuries.
This network shows how religions (blue) are distributed across locations (pink). Circle size represents the total population of Iranian origin in a city or religious followers of Iranian origin. Line thickness shows the strength of each religion's presence in a specific place. This Gephi-generated image connects different religious belongings to the cities of the Red Sea region where Iranians resided during the 19th and early 20th centuries. Drawing upon a database collecting the life trajectories of 128 individuals based on Persian-language sources, the image shows that primary sources tend to overrepresent religious minorities (Baha'is, Jews, Christians, and Sunni Muslims), while underrepresenting the majority religious group (Shia Muslims), as also suggested by the large amount of people whose religious belonging is not specified. Moreover, it distinguishes two different conditions of presence/absence of religious minorities on the two shores of the Red Sea, putting into question the historiographical debate on Jedda's and Aden's cosmopolitanism before the end of empires.